

Q.34 Write the factors which affect the selection of materials.

Q.35 A line shaft rotation at 250r.p.m. is to transmit 25KW. The shaft is assumed to be made of mild steel which allowable shear stress of 40 N/mm^2 . Calculate the diameter of the shaft.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What is stress Concentration? Explain the method of reducing Stress Concentration.

Q.37 A cam is to give the following motion to a knife edged follower:

- a) Outstroke during 60° of Cam rotation
- b) Dwell for the next 30° of Cam rotation
- c) Return stroke during next 60° of Cam rotation.
- d) Dwell for the remaining 210° of Cam rotation.

The stroke of follower is 50mm and minimum Radius for the cam is 60mm.

The followers moves with uniform velocity motion during outstroke and return stroke. Draw the profile of Cam.

Q.38 Explain Gear nomenclature with diagram.

No. of Printed Pages : 4
Roll No.

MSIL-121745/031745

4th Sem / Mech. Engg. (MSIL) Sub : Machine Design & Drawing

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is a brittle material

- a) Aluminium b) Nickel
- c) Platinum d) Cast Iron

Q.2 The deformation per unit length is called

- a) Strain b) Stress
- c) Elasticity d) Plasticity

Q.3 In case of hypoid gears. The mutual position of shaft is

- a) Parallel
- b) Non parallel
- c) Intersecting
- d) Non parallel Non intersecting

Q.4 The size of gear is usually specified by

- a) Circular Pitch b) Pitch Circle diameter
- c) Diametral Pitch d) Pressure angle

Q.5 Practical applications of Cam are in

- a) Gear cutting machines
- b) Printing machines

- c) Sewing machines
- d) All of the above
- Q.6 The period in which there is no movement of follower for Cam rotation is called
 - a) Dwell
 - b) Ascent
 - c) Descent
 - d) None of the above
- Q.7 A bench vice is provided with
 - a) Acme threads
 - b) Buttress threads
 - c) B.S.W. threads
 - d) Square threads
- Q.8 Which thread is used in lead screw of lathe?
 - a) B.S.W.
 - b) Square
 - c) B.A.
 - d) Acme
- Q.9 The standard length of shaft is
 - a) 5m
 - b) 6m
 - c) 7m
 - d) All of the above
- Q.10 The ratio of ultimate stress and working stress is called
 - a) Modulus or Rigidity
 - b) Factor of safety
 - c) Bulk Modulus
 - d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Strain
- Q.12 Stress Concentration.
- Q.13 Screw Thread.

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- Q.14 Toughness.
- Q.15 Crest
- Q.16 Diametral pitch.
- Q.17 Dedendum of gear.
- Q.18 Shear stress.
- Q.19 Radial Cam.
- Q.20 Module.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is the difference between a design work and an undesigned work?
- Q.22 Draw stress strain diagram for mild steel.
- Q.23 Write the general procedure for designing the new machine.
- Q.24 Explain the maximum shear stress theory.
- Q.25 Explain taper sunk key with diagram.
- Q.26 Explain square thread with diagram.
- Q.27 Define any four terms of screw threads.
- Q.28 Explain any two types of Cam with diagram.
- Q.29 Explain any five mechanical properties of materials.
- Q.30 Write the advantages of gear drive.
- Q.31 Find the module and diametral pitch of gear if its pitch circle diameter is 60mm and number of teeth is 20.
- Q.32 Explain Helical gear and Herringbone gear.
- Q.33 Write various function of key.

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